

OpenRocket-Plus: Open-Source Supersonic Aerodynamic Extensions for Rocket Trajectory Simulation

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Summary

OpenRocket-Plus extends OpenRocket (Niskanen 2009) with a physics-based supersonic aerodynamic model suite, enabling six-degree-of-freedom trajectory simulation of high-power and research rockets into the supersonic regime. The original OpenRocket aerodynamic core is built on the Barrowman slender-body method (Barrowman 1967), which assumes subsonic, shock-free flow and becomes progressively inaccurate above approximately Mach 0.8. OpenRocket-Plus adds a shock-geometry pre-pass and a set of compressible-flow component models — shock relations, wave drag, base drag, and compressible skin friction — to the same well-structured Java codebase, while preserving full backward compatibility with existing `.ork` design files and the familiar trajectory simulator. The fork is developed with the goal of eventual integration into upstream OpenRocket.

The accompanying research article (Yu 2026a) reports the scientific validation: integrated flight prediction reaches statistical parity with the commercial baseline RASAero II (Rogers 2015) on an externally selected 25-flight corpus spanning Mach 0.54–4.33. This software paper describes the artifact itself — its architecture, installation, testing, and extensibility — and refers the reader to the companion paper (Yu 2026a) and to the open Rocket Flight Database (Yu 2026b) for the underlying science and ground-truth data.

Statement of Need

High-power rocketry vehicles, university sounding rockets, and amateur research projects routinely reach Mach 2 and beyond. Open-source trajectory simulators — OpenRocket and RocketPy (Ceotto et al. 2021) — rely on the Barrowman equations, which assume subsonic, shock-free, incompressible-boundary-layer flow and therefore lose fidelity through the transonic regime and above. Several specific limitations constrain the original OpenRocket above approximately Mach

0.8:

1. A clamped compressibility factor (`MIN_BETA = 0.25`) that freezes β through the transonic regime instead of following $\beta = \sqrt{|M^2 - 1|}$, flattening every coefficient that scales with $1/\beta$.
2. No wave-drag computation: fin and body pressure drag are missing supersonically, under-predicting total drag.
3. Tabulated nose-pressure data limited to roughly Mach 3, with unphysical linear extrapolation above that.
4. Linear fits for speed of sound and viscosity that err well outside the near-sea-level temperature range.
5. No shock geometry: every component receives freestream conditions even though the nose shock reduces local Mach at downstream stations.
6. No supersonic center-of-pressure correction, with consequences for predicted stability margins.

The commercial tool RASAero II (Rogers 2015) offers the needed accuracy for this class of work but is closed-source, Windows-only, and cannot be extended or independently audited. Recent open-source aerospace tooling (Quintart et al. 2025) and amateur-supersonic studies underscore the demand for inspectable, reproducible simulation, yet no open-source tool currently couples six-degree-of-freedom trajectory simulation with validated supersonic aerodynamics. OpenRocket-Plus fills that gap.

State of the Field

Table 1 summarizes the aerodynamic capabilities of the principal tools.

Table 1. Aerodynamic capability of comparable simulation tools.

Tool	Trajectory	Supersonic aero	Open source
OpenRocket (original)	Yes (6-DOF)	No (Barrowman, $M < 0.8$)	Yes
RASAero II	Yes (3-DOF)	Yes	No
RocketPy	Yes (6-DOF)	No (Barrowman)	Yes
Digital DATCOM	No	Yes (empirical)	No
OpenFOAM / SU2	No	Yes (CFD)	Yes
OpenRocket-Plus	Yes (6-DOF)	Yes	Yes

Missile DATCOM and the Aeroprediction lineage (Sooy and Schmidt 2005) provide many of the same empirical methods but require specialist expertise and produce coefficient tables rather than integrated trajectories. Full CFD codes such as OpenFOAM or SU2 resolve supersonic flows in detail but are impractical for iterative vehicle design or routine flight simulation. OpenRocket-Plus uniquely combines six-degree-of-freedom trajectory simulation, supersonic aerodynamics, and a fully open, extensible codebase.

Software Design

OpenRocket-Plus is written in Java and built within OpenRocket’s multi-module Gradle project (a core simulation library plus a Swing UI). The aerodynamic pipeline follows the original Barrowman pattern — an orchestrating `BarrowmanCalculator` delegates to per-component calculators — extended by a supersonic pre-pass:

```
BarrowmanCalculator
+-- ShockGeometry.compute()      [supersonic pre-pass, inert at M < 1]
+-- BarrowmanStabilityCalculator [CN, CP, Cmq, Magnus, vortex sideforce]
+-- BarrowmanDragCalculator      [friction + pressure + base + power-on]
```

`ShockGeometry` runs once per timestep. It walks the rocket nose-to-tail, applying the θ – β – M relation and Taylor–Maccoll cone flow (Taylor and Maccoll 1933) at the nose, then marching downstream through transitions with oblique-shock and Prandtl–Meyer relations (Ames Research Staff 1953) to produce local Mach, pressure, and temperature at each axial station. The pre-pass is verified bit-for-bit against Taylor–Maccoll cone solutions and analytic shoulder geometry. Below Mach 1 it returns freestream conditions with no extra computation. Each downstream calculator then reads its local post-shock conditions rather than freestream, correcting fin normal-force slopes, dynamic pressure, interference factors, and drag coefficients. The pre-pass is best understood as an architectural seam that supplies correct local flow to the supersonic component models; its contribution to integrated apogee is small (the dominant apogee driver is base drag), but its local-flow fidelity is what makes the downstream supersonic models physically consistent (Yu 2026a).

The principal component models are: a complete oblique/normal-shock and Prandtl–Meyer package (Ames Research Staff 1953); Taylor–Maccoll exact wave drag for conical noses (Taylor and Maccoll 1933) with shock-expansion theory for ogives; DATCOM Section 4.1.5.1 fin wave drag (Finck 1979); a base-drag suite spanning laminar, turbulent, and boattail regimes whose supersonic correlation ($C_{d,\text{base}} = 0.064 + 0.186/M^2$) is presented as an empirical correlation validated against NACA TN 3393 (Reller and Hamaker 1955) and consistent with ESDU 77021 (Engineering Sciences Data Unit 1977); Van Driest II compressible skin friction (Hopkins and Inouye 1971); and Modified Newtonian theory for the hypersonic regime (Anderson 2006).

All Mach-regime transitions are C^1 -continuous. The compressibility factor $\beta = \sqrt{|M^2 - 1|}$ uses a cubic Hermite spline through M 0.95–1.05; base drag blends across M 0.85–1.3; turbulent base drag activates over M 1.2–1.4; fin wave drag ramps in over M 0.9–1.2; Modified Newtonian blends into the shock-expansion result over M 4.0–6.0. C^1 -continuity is an architectural requirement: discontinuities cause the Runge–Kutta trajectory integrator to oscillate when the vehicle repeatedly crosses a Mach threshold.

Table 2. Mach-regime blending regions ensuring C^1 -continuous simu-

lation.

Physical quantity	Blending range	Method
Compressibility factor β	M 0.95–1.05	Cubic Hermite spline
Skin friction	M 0.9–1.1	Polynomial blend
Base drag	M 0.85–1.3	C^1 cubic blend
Fin wave drag	M 0.9–1.2	Hermite activation
Modified Newtonian	M 4.0–6.0	Smoothstep

An alternative `LookupTableDragCalculator` and `LookupTableStabilityCalculator` accept user-supplied CSV tables as overrides, enabling direct comparison against CFD or wind-tunnel data for a specific vehicle.

Installation and Use

OpenRocket-Plus builds with the bundled Gradle wrapper and requires a Java 17+ JDK. The graphical application launches with `./gradlew run`; a runnable distribution is produced with `./gradlew :swing:build`. Existing OpenRocket `.ork` design files open unchanged, and the supersonic models engage automatically once a simulation crosses into the transonic regime — no configuration is required for the default physics path. Per-vehicle CSV override tables can be supplied through the lookup-table calculators for users who wish to drive the trajectory with external CFD or wind-tunnel coefficients. Full build and usage instructions are maintained in the repository README.

Testing

Each aerodynamic model is an independently testable Java class with an explicit Mach validity range, and the supersonic suite is covered by a dedicated set of JUnit tests (the broader project ships a large existing OpenRocket test suite). The fast core test tier runs in roughly half a minute via `./gradlew :core:test`; heavier statistical and parameter-sweep tiers are opt-in through Gradle properties. Each model carries claim-by-claim checks against a primary reference, and open heuristic and geometry-family gaps are explicitly classified rather than silently treated as closed.

Table 3 lists representative component benchmarks; the metrics, references, and their honest interpretation are documented in full in the companion research article (Yu 2026a).

Table 3. Representative component-level benchmarks.

Subsystem	Reference	Metric	Value
Oblique/normal-shock relations	NACA 1135 (Ames Research Staff 1953)	max err	< 0.01%
Taylor–Maccoll cone	Cone tables (Taylor and Maccoll 1933)	max shock-angle rel err	0.825%
Nose/body wave drag (5 shapes)	NACA RM A52H28 (NACA Ames Aeronautical Laboratory 1952)	MAE	0.029
Base drag, turbulent	NACA TN 3393 (Reller and Hamaker 1955)	MAPE	15.9%
Base drag, laminar (Chapman)	NACA TN 3393 (Reller and Hamaker 1955)	MAPE	4.4%
Fin wave drag (DATCOM 4.1.5.1)	NACA TN 3650 (Welsh 1956)	MAPE	21%
Fin $C_{N\alpha}$ (M 0.6–5.82)	NASA TM X-653 (Nielsen et al. 1962)	MAPE	6.84%
Fin x_{CP}/d (M 0.6–5.82)	NASA TM X-653 (Nielsen et al. 1962)	MAPE	7.11%
Basic Finner total drag (M 1.08–4.30)	ADA636861 (Dupuis and Hathaway 1997)	MAPE	11.8%
Hypersonic cone foredrag (M 6.5–17.2)	DTIC AD0487365 (Grabow 1965)	MAPE	19.7%

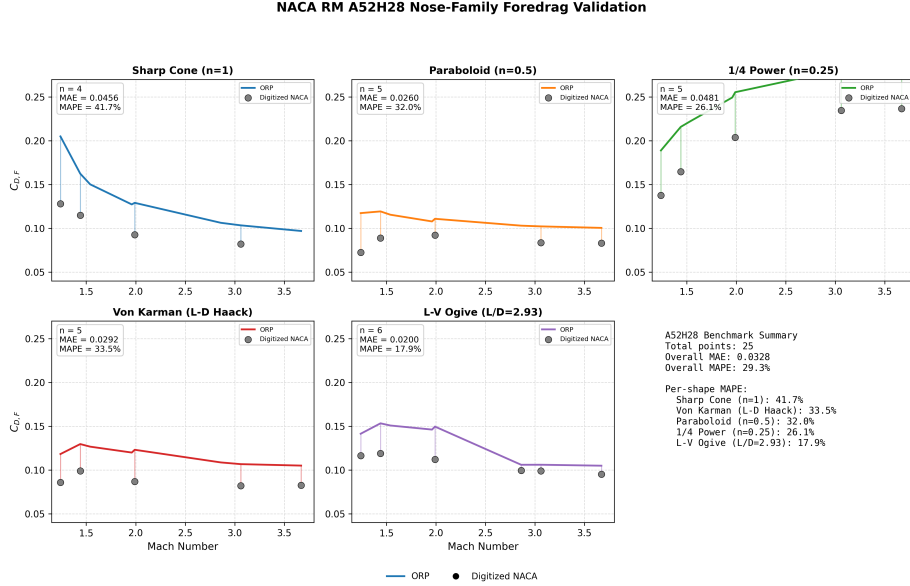


Figure 1: Nose/body wave-drag coefficient versus Mach number for five nose shapes, comparing OpenRocket-Plus against NACA RM A52H28 wind-tunnel data. The headline accuracy metric is the aggregate mean absolute error, $MAE = 0.029$; the corresponding MAPE of 29.3% is inflated by the small absolute drag values of the power-law and Haack shapes and is reported here only for context.

The shock solver matches NACA 1135 tabular values to better than 0.01% across the supersonic range. Nose/body wave drag is validated on five distinct nose shapes with an aggregate MAE of 0.029 (the headline metric); the corresponding MAPE of 29.3% is a small-denominator artifact for the low-drag power-law and Haack shapes and is not used as the headline figure. Fin aerodynamics validated against NASA TM X-653 (M 0.6–5.82) give $C_{N\alpha}$ MAPE 6.84% and center-of-pressure MAPE 7.11%. The Basic Finner free-flight range benchmark (total-drag MAPE 11.8%, M 1.08–4.30) (Dupuis and Hathaway 1997) is a stringent test because it includes manufacturing variation, sabot effects, and base-flow unsteadiness absent from wind-tunnel pressure data. The hypersonic cone-foredrag benchmark (MAPE 19.7%, 11 points, M 6.5–17.2) (Grabow 1965) establishes the upper Mach extent of *component-level* validation.

Scope of Validation

The companion research article (Yu 2026a) validates integrated flight prediction to **Mach 4.33** on an externally selected 25-flight corpus (Mach 0.54–4.33; released as the Rocket Flight Database (Yu 2026b) under CC-BY-4.0), reporting

statistical parity with the version-locked RASAero II baseline — parity, **not** superiority. See that article for the full statistics, hypothesis test, and confidence intervals.

Component benchmarks extend higher in Mach than the integrated flight set: the cone-foredrag check reaches **Mach 17.2** (Table 3). Separately, an **exploratory** high-Mach demonstration runs roughly twenty historical sounding-rocket flights and shows that the method *can reach* Mach 7 within $\pm 7\%$ on well-characterized vehicles (e.g., Black Brant VB at Mach 7.22), while motor- and geometry-reconstruction uncertainty dominates on poorly documented historical flights. That exploratory set is reported in full in (Yu 2026a) and is explicitly **not** a validation headline; OpenRocket-Plus is presented as flight-validated to Mach 4.33 and component-validated to Mach 17.2.

Limitations

Documented limitations include: the transonic band (M 0.8–1.3), the most challenging regime for all comparable tools and the disclosed weakness here; elevated MAPE for the low-drag power-law and Haack noses, where small absolute values amplify percentage error (the MAE of 0.029 is the meaningful metric); pitch-damping (C_{mq}) dynamic-stability derivatives, held at a lower confidence tier (the transonic augmentation overshoots reference CFD), affecting predicted coning but not apogee; and high-fineness slender-body pressure drag that decays toward zero above Mach 5, producing a quantified positive apogee bias on high-L/D hypersonic flights (documented as ongoing work). Real-gas dissociation chemistry above roughly Mach 7 is not modeled. The full statistical treatment, including the in-sample disclosure for two base-drag scale constants and a decontaminated prospective holdout demonstrating generalization, is given in the companion article (Yu 2026a).

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